Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	11
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	10.3636	10	3.2641	count
ring_length_r30_px	headline	423.091	442	90.8366	px
ring_length_r60_px	reported	989.091	998	188.498	px
owned_length_px	ownership-dependent	2954.87	2421.54	1411.64	px
edge_count	ownership-dependent	138.273	103	69.933	count
mean_edge_length_px	ownership-dependent	21.6005	21.4048	1.8936	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22488.8	22056	3166.3	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0478154	0.0490921	0.00947425	px^-1
ring_density_r60	normalised	0.0442888	0.0452485	0.00791038	px^-1
roots_per_100px_perimeter	normalised	4.6662	4.6282	1.0772	per 100 px
ring_attached_length_r30_px	network v2	389.818	399	93.1051	px
ring_attached_length_r60_px	network v2	824.182	828	220.279	px
ring_length_w_r30_px	network v2	479.951	514.316	101.751	px
ring_length_w_r60_px	network v2	1119.82	1128.31	211.225	px
sholl_crossings_r10	network v2	10.1818	10	2.6389	count
sholl_crossings_r20	network v2	12.6364	13	2.6181	count
sholl_crossings_r30	network v2	13.8182	13	3.2193	count
loop_count	loops, report only	13.0909	10	10.2123	count
loop_edge_share	loops, report only	0.3429	0.3495	0.1099	share
ring_width_mean_r30_px	width	6.8505	6.952	0.5632	px

Per field
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measure	value	units
analysed_area_px2	1.02271e+06	px^2
total_skeleton_length_px	41329	px
canalicular_length_density_per_px	0.0404112	px^-1
junction_count	987	count
junction_density_per_px2	0.00096508	px^-2
lacuna_count	14	count

lacunae_per_px2	1.369e-05	px^-2
skeleton_component_count	242	count
mean_component_length_px	170.781	px
median_component_length_px	16.5	px
field_length_density_w_per_px	0.0457258	px^-1
field_density_without_flagged_per_px	0.0401152	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	530	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0404112	px^-1
field_length_density_with_unattached_per_px	0.0409294	px^-1
loop_count	218	count
loop_edge_share	0.3908	share
edge_count	2142	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8295	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	38500	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	448	43.7126	19.8458	2.2026	0.891	0.6767	1.435	7.0045	363.067	True
3	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
4	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
5	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
6	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
7	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
8	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
9	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
10	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
11	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
12	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
13	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
14	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

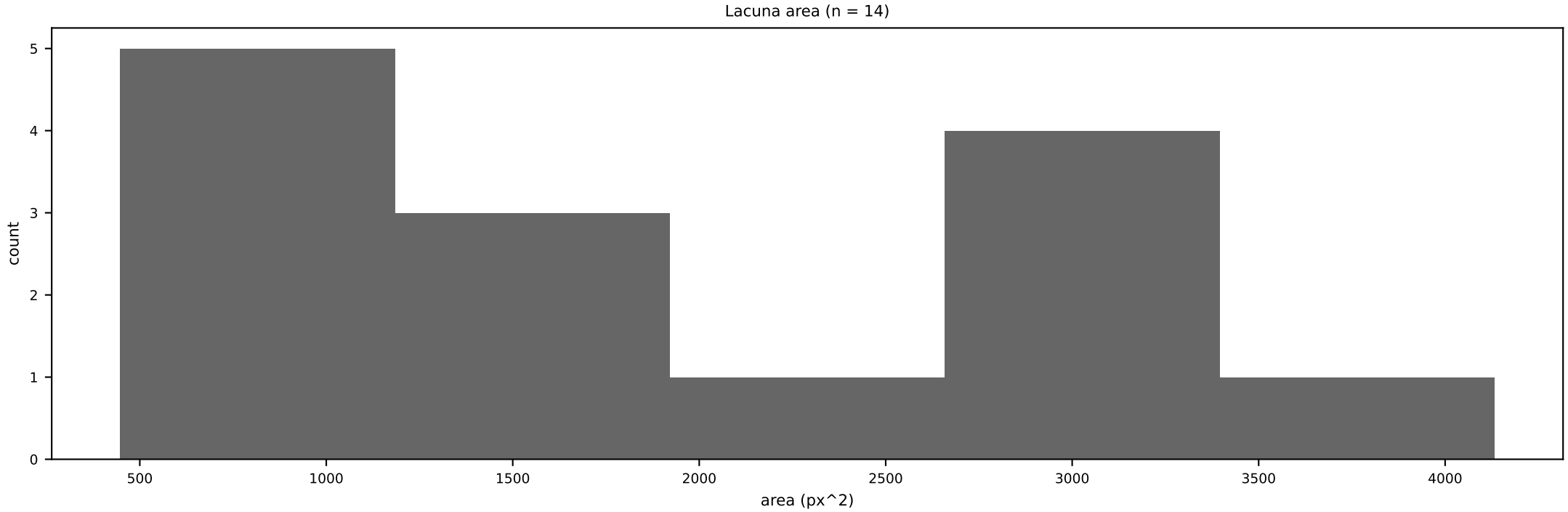
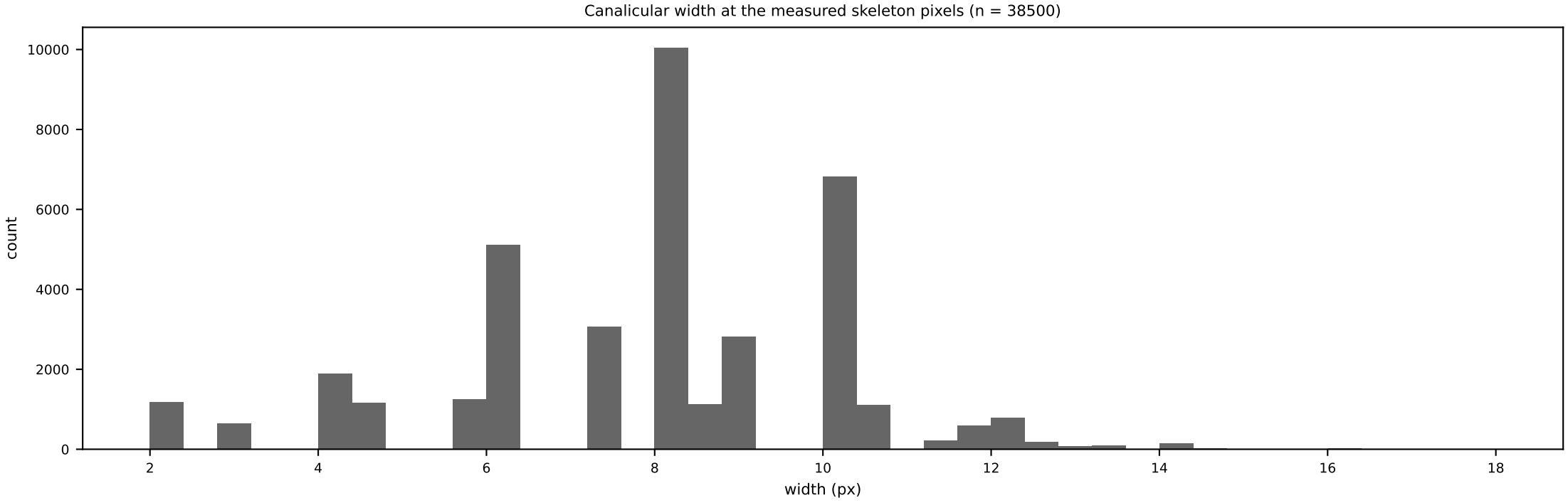
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	160	449	3147.48	160	19.6717	
2	4	77	223	1381.47	78	17.7111	
3	8	400	825	1845.5	95	19.4263	6.0035
4	11	482	1109	5491.25	276	19.8958	6.952
5	16	537	1190	4775.42	216	22.1084	7.3889

6	5	421	1046	4046.57	198	20.4372	7.6867
7	13	465	998	4104.65	193	21.2676	6.8683
8	10	460	980	1999.21	101	19.7941	7.4533
9	10	351	815	1097.88	52	21.1131	7.4606
10	10	546	1365	2421.54	103	23.5101	7.0158
11	7	311	739	2780.9	107	25.9898	6.7399
12	14	442	1011	1505.18	68	22.135	6.0426
13	8	267	790	1391.31	65	21.4048	6.9734
14	12	323	827	2142.01	99	21.6365	6.2309

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0432316	0.0441886	3.8039
2	132.847	3699	7686	0.6031	0.5656	0.0208164	0.0290138	3.011
3	219.681	8976	20715	1	0.9689	0.0445633	0.0398262	3.6416
4	210.244	8511	22591	1	1	0.0566326	0.0490903	5.232
5	335.421	12178	29932	1	1	0.0440959	0.0397568	4.7701
6	167.238	7485	20582	1	1	0.0562458	0.0508211	2.9898
7	243.522	9472	22056	1	1	0.0490921	0.0452485	5.3383
8	167.924	7349	17555	1	1	0.0625935	0.0558246	5.9551
9	282.894	8678	19332	0.813	0.7462	0.0404471	0.0421581	3.5349
10	263.48	10172	24552	1	1	0.0536768	0.0555963	3.7954
11	174.51	7570	20770	1	1	0.0410832	0.0355802	4.0112
12	210.167	8646	22189	1	1	0.0511219	0.0455631	6.6614
13	185.823	7980	21610	1	1	0.0334586	0.0365571	4.3052
14	259.279	9669	24825	1	1	0.0334057	0.0333132	4.6282

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	145	372	175.468	498.019	5	5	6
2	70	158	87.0122	245.024	2	1	3
3	314	530	444.291	911.34	10	12	15
4	477	1084	554.588	1272.64	13	13	12
5	508	942	597.203	1326.49	10	17	15
6	399	841	493.446	1207.89	11	13	13
7	439	828	524.475	1128.31	12	14	12
8	448	951	520.162	1099.67	11	13	15
9	259	547	400.191	919.21	9	12	11
10	499	1228	611.546	1536.79	15	16	21
11	267	627	351.664	853.394	6	12	10
12	384	825	514.316	1148.35	9	12	17
13	257	643	306.179	908.365	8	9	11
14	296	567	361.593	924.826	7	8	11

682_z23: distributions, pre-validation, px